Mating Connector:
Housing: Molex 0436450200
Precrimped 18AWG wire 450mm:
Black: Molex 2147601225
Red: Molex 2147602225

Reverse Polarity Protection

Q9
CSD19534Q5A

J10
POWER
V_IN
V_IN_RTN
Molex
0436500200

D1
SMDJ24CA
24V

C54
10uF-50V-10%-X7R-1210

C52
0.1uF-50V-10%-0603

U7
LM74700QDBVRQ1
VCAP
EN
GND
A
G
C

C45
10uF-50V-10%-X7R-1210

R28
5.1k-1%-0603

R27
5.1k-1%-0603

R26
5.1k-1%-0603

D2
PWR

VIN_PS

File: buck-boost-power-supply.kicad_sch
2/5
+V_IN
-V_IN
15V 3A Buck-Boost Power Supply
+V_OUT
-V_OUT

Power Supply Back
Feed Protection

Q1
CSD19534Q5A

C1
10uF-50V-10%-X7R-1210

C11
0.1uF-50V-10%-0603

U1
LM74700QDBVRQ1
VCAP
EN
GND
A
G
C

C12
10uF-50V-10%-X7R-1210

J4
V_OUT
V_OUT_RTN
15V
Molex
0436500200

Mating Connector:
Housing: Molex 0436450200
Precrimped 18AWG wire 450mm:
Black: Molex 2147601225
Red: Molex 2147602225

Mating Connector:
Housing: Molex 5040510401
Precrimped 24AWG wire 150mm: Molex 0797581003

J7
USB1
Molex
5040500491

File: canable2.kicad_sch
3/5
CANable2.0
USB_DP
USB_DM
3V3_USB
GND_CAN
CAN_HC
CAN_LC

J9
CAN1
MFG P/N: 0705550036
MFG: Molex

J1
Terminal
J2
Terminal
Keystone Electronics
7774

J3
NX CONTROL & UART
3V3_BTN
NX_PC_LED-
NX_PC_LED+
UART2_RXD
UART2_TXD
AC OK
Auto Power On Disable
SYS_RESET
REC
GND
SLEEP/WAKE*
Molex
PH1RB-15-UA
GND1

3V3_BTN
UART2_TXD
UART2_RXD
SYS_RESET
REC
NX_PC_LED+
NX_PC_LED-

D3
PC

SW2
RECOVERY

SW1
RESET

GND1

File: NX-Button-Interface.kicad_sch
4/5
NX-Button Interface

File: usb-serial.kicad_sch
5/5
USB Serial
3V3
RXD
TXD
RESET
FORCE_RECOVERY
UART2_TXD
UART2_RXD

H1
MountingHole
H5
MountingHole
H6
MountingHole
H2
MountingHole
H7
MountingHole
H8
MountingHole
H9
MountingHole

H10
1.152mm ToolingHole
H11
1.152mm ToolingHole
H12
1.152mm ToolingHole
H13
1.152mm ToolingHole

Light Pipe
LP1
Bivar
Bivar-SLP3-150-100-F
LP2
Bivar
Bivar-SLP3-150-100-F
LP3
Bivar
Bivar-SLP3-150-100-F
LP4
Bivar
Bivar-SLP3-150-100-F

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Sheet: /
File: TVS-Adapter-v1.kicad_sch

Title: The Vision System Adapter

Size: B	Date: 2026-10-14	Rev: 1
KiCad E.D.A. 10.0.7		Id: 1/5

+V_IN
−V_IN
+V_OUT
−V_OUT
GND
Vcc

C59 47uF 50V
C58 47uF 50V
C44 10uF−50V−10%−X7R−1210
C9 10uF−50V−10%−X7R−1210
C6 10uF−50V−10%−X7R−1210
C8 10uF−50V−10%−X7R−1210
C10 10uF−50V−10%−X7R−1210
C43 10uF−50V−10%−X7R−1210
C49 10uF−50V−10%−X7R−1210
C50 10uF−50V−10%−X7R−1210
C7 10uF−50V−10%−X7R−1210

HDRV1 Q8 CSD19534Q5A
LDRV1 Q6 CSD19534Q5A
Q3 CSD17578Q5A HDRV2
Q4 CSD17578Q5A LDRV2
SW1 L1 SW2
2.2uH−19A−5.1mohm−20%

RCS
R14 10mOhm−3W−1%−2512
CSG

C15 10uF−50V−10%−X7R−1210
C19 10uF−50V−10%−X7R−1210
C18 10uF−50V−10%−X7R−1210
C14 10uF−50V−10%−X7R−1210
C24 10uF−50V−10%−X7R−1210
C23 10uF−50V−10%−X7R−1210
C2 10uF−50V−10%−X7R−1210
C4 10uF−50V−10%−X7R−1210
C5 10uF−50V−10%−X7R−1210
C3 10uF−50V−10%−X7R−1210

D8 BAT54KFILM
R21 249K−1%−0603
R19 93.1K−1%−0603
R17 2K−1%−0603
R20 10R0−1%−0603
C39 0.1uF−50V−10%−0603

Texas Instruments
LM5176PWPR
U4
VIN
VISNS
EN/UVLO
SS
COMP
MODE
RT
DITH
SLOPE
CS
CSG
VCC
BIAS
VOSNS
HDRV1
BOOT1
SW1
LDRV1
FB
PGOOD
HDRV2
BOOT2
SW2
LDRV2
ISNS(+)
ISNS(−)
AGND
PGND
EP

C37 0.1uF−50V−10%−0603
R3 357K−1%−0603
R5 20K−1%−0603
D7 BAT54KFILM
C36 1uF−50V−10%−0603
C40 0.1uF−50V−10%−0603
D6 BAT54KFILM
C29 0.1uF−50V−10%−0603
R22 0R−1%−0603
R18 0R−1%−0603
R10 0R−1%−0603
R13 0R−1%−0603

C33 15nF−10%−50V−0603
C26 5.6nF−10%−50V−0603
R6 5.9K−1%−0603
C28 5.6nF−10%−50V−0603
C31 270pF−10%−50V−0603
R16 205K−1%−0603
R15 13K−1%−0603
C35 150pF−50V−10%−0603
C27 47pF−50V−10%−0603
R8 100R−1%−0603
R7 100R−1%−0603

VIN(MAX)	24V
VIN(MIN)	5V
VOUT	15V
IOUT(MAX)	3A
FSW	589 kHz
Efficiency	96.7%

15V−3A−Power Supply Analysis

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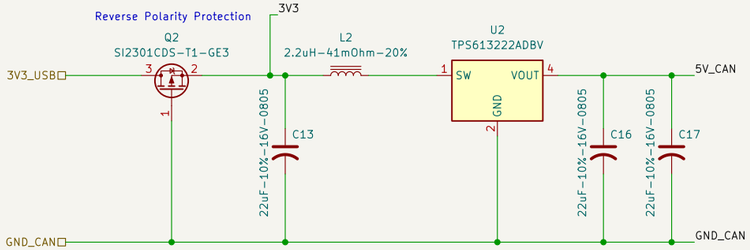
Sheet: /15V 3A Buck−Boost Power Supply/
File: buck−boost−power−supply.kicad_sch

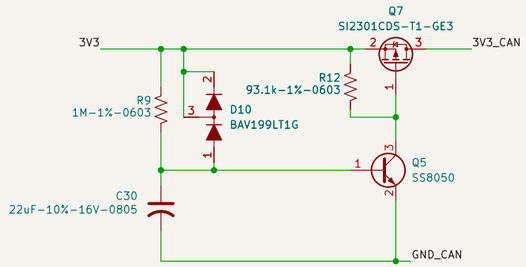
Title: The Vision System Adapter

Size: B	Date: 2026−10−14	Rev: 1
KiCad E.D.A. 10.0.7		Id: 2/5

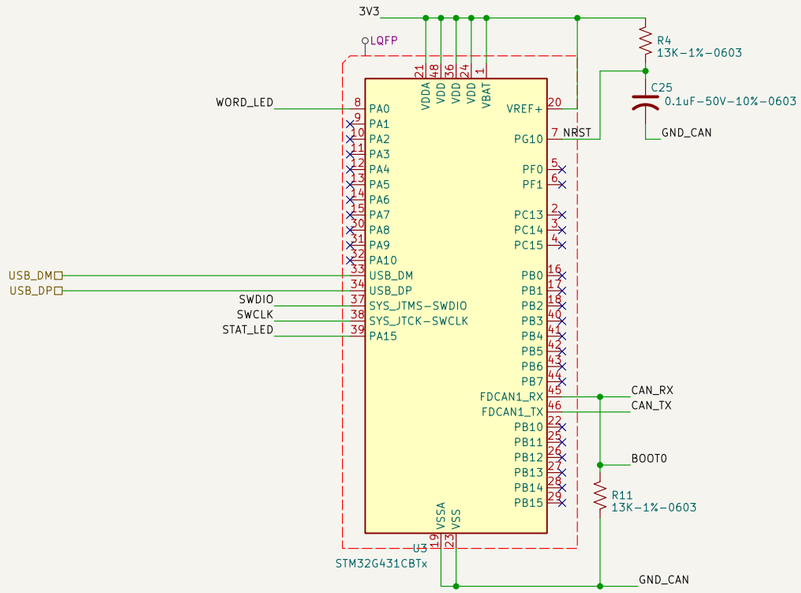
5V-250mA-Power Supply Analysis

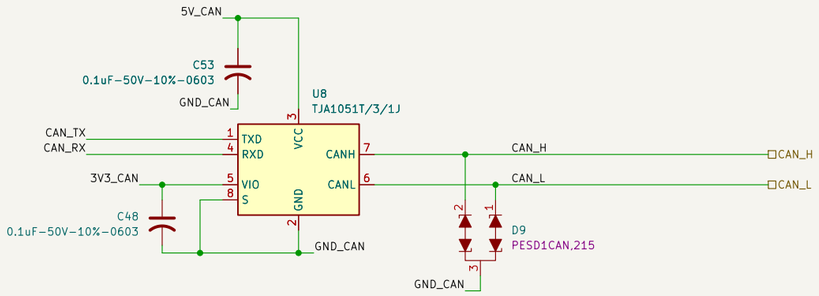
Reverse Polarity Protection

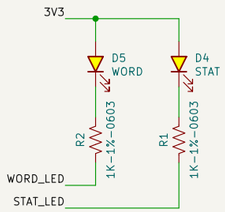




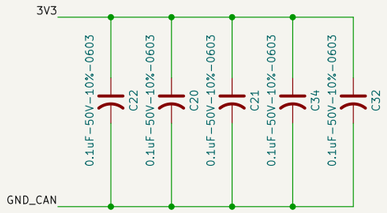
3V3_CAN delays bringing CAN RX (BOOT0) high until NPN turns on. ~10sec.











CANable 2.0 Schematic

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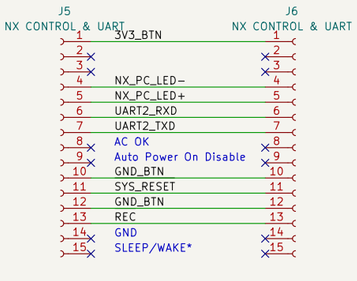
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Sheet: /CANable2.0/
File: canable2.kicad_sch

Title: The Vision System Adapter

Size: B	Date: 2026-10-14	**Rev: 1**
KiCad E.D.A. 10.0.7		Id: 3/5



H3
M3.7 MountingHole

H4
M3.7 MountingHole

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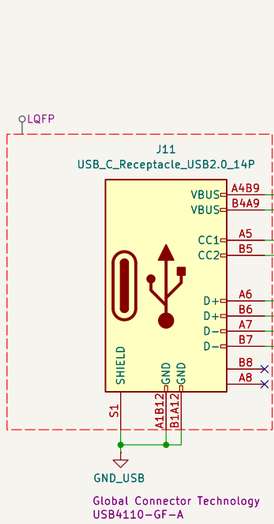
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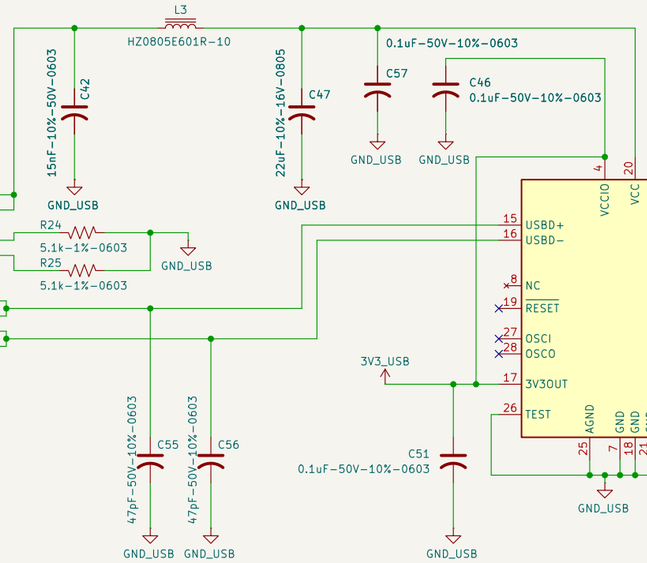
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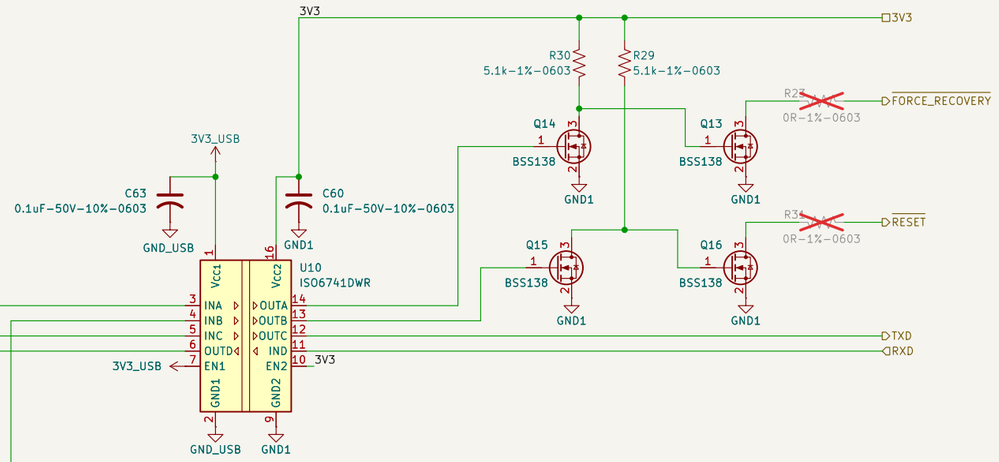
Sheet: /NX-Button Interface/
File: NX-Button-Interface.kicad_sch

Title: The Vision System Adapter

Size: B	Date: 2026-10-14	**Rev: 1**
KiCad E.D.A. 10.0.7		Id: 4/5







https://hackaday.com/2023/08/07/all-about-usb-c-example-circuits/

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Sheet: /USB Serial/
File: usb-serial.kicad_sch

Title: The Vision System Adapter

Size: B	Date: 2026-10-14	**Rev: 1**
KiCad E.D.A. 10.0.7		Id: 5/5